



BunkerHill Health
% John Smith, JD Partner
Hogan Lovells US, L.L.P.
555 Thirteenth Street NW,
WASHINGTON, DC 20004

July 1, 2025

Re: K243779

Trade/Device Name: Bunkerhill Abdominal Aortic Quantification (AAQ)
Regulation Number: 21 CFR 892.2050
Regulation Name: Medical Image Management And Processing System
Regulatory Class: Class II
Product Code: QIH
Dated: June 2, 2025
Received: June 2, 2025

Dear John Smith:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device"

(<https://www.fda.gov/media/99812/download>) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (<https://www.fda.gov/media/99785/download>).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system>.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the

Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

A handwritten signature in black ink that reads "Jessica Lamb". The signature is written in a cursive style. A large, light blue "FDA" watermark is visible in the background behind the signature.

Jessica Lamb
Assistant Director
Imaging Software Team
DHT8B: Division of Radiologic Imaging
Devices and Electronic Products
OHT8: Office of Radiological Health
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

Submission Number (if known)

K243779

Device Name

Bunkerhill Abdominal Aortic Quantification (AAQ)

Indications for Use (Describe)

Bunkerhill AAQ is a radiological image processing system software indicated for use in the analysis of CT exams with or without contrast, that include the L1 – L5 region of the abdominal aorta, in adults aged 22 and older.

The device is intended to assist appropriately trained medical specialists by providing the user with the maximum axial abdominal aortic axial diameter measurement of cases that include the abdominal aorta. Bunkerhill AAQ is indicated to evaluate normal and aneurysmal abdominal aortas and is not intended to evaluate post-operative aortas.

The Bunkerhill AAQ results are not intended to be used on a stand-alone basis for clinical decision-making or otherwise preclude clinical assessment of cases. These measurements are unofficial, are not final, and are subject to change after review by a qualified interpreting physician. For final clinically approved measurements, please refer to the official radiology report. Clinicians are responsible for viewing full images per the standard of care.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

This section applies only to requirements of the Paperwork Reduction Act of 1995.

DO NOT SEND YOUR COMPLETED FORM TO THE PRA STAFF EMAIL ADDRESS BELOW.

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510(K) SUMMARY
Bunkerhill AAQ
K243779

Bunkerhill, Inc.
436 Bryant Street
San Francisco CA 94107

Phone: (408) 8028146
Contact Person: Eren Alkan
Date Prepared: July 1, 2025

Proposed Device

Proprietary Name	Bunkerhill Abdominal Aortic Quantification (AAQ)
Classification Name	Medical image management and processing system
Regulation Number	21 CFR 892.2050
Product Code	QIH
Regulatory Class	II

Predicate Device

Proprietary Name	Briefcase Quantification Device
Premarket Notification	K230534
Classification Name	Medical image management and processing system
Regulation Number	21 CFR 892.2050
Product Code	QIH
Regulatory Class	II

Device Description

Bunkerhill AAQ is a software-only medical device that employs deep learning algorithms to provide automatic maximal abdominal aortic diameter measurements from axial CT scans of the abdomen/pelvis, with or without IV contrast.

Bunkerhill AAQ receives DICOM instances and processes them chronologically by running the algorithm on relevant series to measure the maximum abdominal aortic diameter. Following the AI processing, the output of the algorithm analysis is transferred to standard radiology image review and reporting software.

Bunkerhill AAQ produces a preview image annotated with the maximum axial diameter measurement. The diameter marking is not intended to be a final output, but serves the purpose of visualization and measurement. The original, unmarked series remains available in the PACS as well.

The preview image presents an unofficial and not final measurement, and the user is instructed to review the full image and any other clinical information before making a clinical decision. The image includes a disclaimer: “Not for diagnostic use. The measurement is unofficial, not final, and must be reviewed by a qualified interpreting physician”.

Intended Use / Indications for Use

Bunkerhill AAQ is a radiological image processing system software indicated for use in the analysis of CT exams with or without contrast, that include the L1 – L5 region of the abdominal aorta, in adults aged 22 and older.

The device is intended to assist appropriately trained medical specialists by providing the user with the maximum axial abdominal aortic diameter measurement of cases that include the abdominal aorta. Bunkerhill AAQ is indicated to evaluate normal and aneurysmal abdominal aortas and is not intended to evaluate post- operative aortas.

The Bunkerhill AAQ results are not intended to be used on a stand-alone basis for clinical decision-making or otherwise preclude clinical assessment of cases. These measurements are unofficial, are not final, and are subject to change after review by a qualified interpreting physician. For final clinically approved measurements, please refer to the official radiology report. Clinicians are responsible for viewing full images per the standard of care.

A table comparing the intended use of the subject and predicate devices is provided below.

	Proposed Device: Bunkerhill AAQ Algorithm	Predicate Device: Briefcase Quantification (K230534)
Intended use / Indications for use	Bunkerhill AAQ is a radiological image processing system software indicated for use in the analysis of CT exams with or without contrast, that include the L1 – L5 region of the abdominal aorta, in adults aged 22 and older. The device is intended to assist appropriately trained medical specialists by providing the user with the maximum axial abdominal aortic diameter measurement of cases that include the abdominal aorta. Bunkerhill AAQ is indicated to evaluate normal and aneurysmal abdominal aortas and is not intended to evaluate post- operative aortas.	BriefCase-Quantification is a radiological image management and processing system software indicated for use in the analysis of CT exams with contrast, that include the abdominal aorta, in adults or transitional adolescents aged 18 and older. The device is intended to assist appropriately trained medical specialists by providing the user with the maximum abdominal aortic axial diameter measurement of cases that include the abdominal aorta (M-AbdAo) BriefCase-Quantification is indicated to evaluate normal and aneurysmal abdominal aortas and is not intended to evaluate post- operative aortas.

	Proposed Device: Bunkerhill AAQ Algorithm	Predicate Device: Briefcase Quantification (K230534)
	The Bunkerhill AAQ results are not intended to be used on a stand-alone basis for clinical decision-making or otherwise preclude clinical assessment of cases. These measurements are unofficial, are not final, and are subject to change after review by a qualified interpreting physician. For final clinically approved measurements, please refer to the official radiology report. Clinicians are responsible for viewing full images per the standard of care.	The BriefCase-Quantification results are not intended to be used on a stand-alone basis for clinical decision-making or otherwise preclude clinical assessment of cases. These measurements are unofficial, are not final, and are subject to change after review by a radiologist. For final clinically approved measurements, please refer to the official radiology report. Clinicians are responsible for viewing full images per the standard of care.

Summary of Technological Characteristics

At a high level, the subject and predicate devices are based on the following same technological elements:

- Both the predicate and the subject device use deep-learning algorithms to assist the medical professionals with the maximum abdominal aortic axial diameter measurement.
- Both devices analyze non-gated chest computed tomography (CT) images that are sent to the software in DICOM format.
- Both devices serve as support tools to provide information to the physician. Both can be used on- demand or optionally by the physician and do not provide a definitive diagnosis. Both devices do not replace clinical evaluation and do not alter the standard of care. Both require the physician to use this information to decide next steps and/or additional diagnostic work up.
- Both devices provide a preview image annotated with the maximum axial diameter measurement and the original, unmarked series remains available in the PACS.

The following technological difference exists between the subject and predicate devices:

- Minor difference in device input (contrast and non-contrast for subject device vs contrast scans for predicate device)

	Proposed Device: Bunkerhill AAQ Algorithm	Predicate Device: Briefcase Quantification (K230534)	Summary
Product code	QIH	QIH	Same
Regulation number	21 CFR §892. 2050	21 CFR §892. 2050	Same
Modality	Computed tomography (CT)	Computed tomography (CT)	Same
Image format	DICOM	DICOM	Same
Supported CT scan	CT exams with or without contrast that include the abdominal aorta	CT exams with contrast that include the abdominal aorta	Similar
Diameter measurement	Yes	Yes	Same
Algorithm	Artificial intelligence algorithm with database of images.	Artificial intelligence algorithm with database of images.	Same
Interference with standard workflow	No	No	Same
Output	Output can be optionally or on-demand be inserted into the radiology report; produces a preview image annotated with the maximum axial diameter measurement. The original, unmarked series remains available in the PACS as well.	Produces a preview image annotated with the maximum axial diameter measurement. The original, unmarked series remains available in the PACS as well.	Similar

	Proposed Device: Bunkerhill AAQ Algorithm	Predicate Device: Briefcase Quantification (K230534)	Summary
Structure	-Bunkerhill AAQ, is hosted on a cloud server, analyzes applicable CT images that are acquired on CT scanner that are forwarded to Bunkerhill AAQ. - The results of the analysis are exported in DICOM format, and are sent to a PACS destination for review by medical specialists, to assist in the	-BriefCase-Quantification, is hosted on a cloud server, analyzes applicable CT images that are acquired on CT scanner that are forwarded to BriefCase-Quantification. - The results of the analysis are exported in DICOM format, and are sent to a PACS destination for review	Same
	measurement of the abdominal aorta.	by medical specialists, to assist in the measurement of the abdominal aorta.	
Type of Interpretation	Adjunctive information	Adjunctive information	Same
Intended User	Appropriately trained medical specialists	Appropriately trained medical specialists	Same
Patient population	Patients aged 18 years and above	Patients above the age of 18	Same
Anatomical location	Abdominal aorta	Abdominal aorta	Same
Intended location	Medical facility	Medical facility	Same
Rx or OTC	Rx	Rx	Same

Performance Data

Safety and performance of the AAQ algorithm has been evaluated and verified in accordance with software specifications and applicable performance standards through Software Development and Validation & Verification Process to ensure performance according to specifications, User Requirements and Federal Regulations and Guidance documents, “Content of Premarket Submissions for Device Software Functions”.

The AAQ algorithm performance was validated in a stand-alone retrospective study for overall agreement of the device output compared to a ground truth established by 3 U.S. Board Certified Radiologists. The pivotal test set comprised 258 patients sourced North Carolina, Alabama, the greater Washington D.C area, and Sao Paulo, Brazil. The mean-absolute-error = 1.58 mm (95 % CI 1.38–1.80) versus the ≤ 2.0 mm acceptance criterion, and the secondary endpoint with $\Delta ICC = 0.003 < 0.05$, while Bland-Altman limits were $\pm \approx 5$ mm, supporting the characterized performance. The dataset was balanced with 118 male and 140 female patients, the median age of the patient population was 67 years old and the range was 22 years old to 99 years old, the manufacturers represented in the test set were Toshiba, Phillips, Siemens, and GE Healthcare, and the test set consisted of 58% non-contrast studies and 42% contrast studies. The subgroup analyses showed consistent performance across sex, age, aneurysm size, manufacturer, slice-thickness strata, kVp ranges, contrast vs. non-contrast, and collection site demonstrating the generalizability of the algorithm.

Conclusions

The Bunkerhill AAQ algorithm is as substantially equivalent as the predicate Briefcase Quantification device (K230534). The subject device has the same intended uses and similar indications, technological characteristics, and principles of operation as its predicate device. The minor differences in indications do not alter the intended diagnostic use of the device and do not affect its safety and effectiveness when used as labeled. In summary, any minor differences between the AAQ algorithm and the Briefcase Quantification do not raise any issues of substantial equivalence.